

ZHIYUAN QI

Tsinghua University, Department of Electronic Engineering, M.S. in EECS, Beijing, China

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RESEARCH INTERESTS

LLM Agent, VLM/MLLM, 3D Computer Vision, World Model, Spatial Intelligence, VR/AR/XR

EDUCATION

Tsinghua University, M.S. in EECS

GPA:3.73/4.00, 2024 - present

Beijing JiaoTong University, B.E. in Communication Engineering

GPA:3.89/4.00, 2020 - 2024

PUBLICATIONS

1. **Zhiyuan Qi***, Yulong Feng and Zhijin Qin, “Adaptive Sampling and Joint Semantic-Channel Coding Under Dynamic Channel Environment,” **ICC 2025** - IEEE International Conference on Communications, Montreal, QC, Canada, 2025, pp. 518-523, doi: 10.1109/ICC52391.2025.11161632.
2. **Zhiyuan Qi***, Jingkai Ying*, Yulong Feng, Zhijin Qin, Zhu Han, Rahim Tafazolli and Yonina C Eldar, “Semantic Communication Enabled Holographic Video Processing and Transmission,” IEEE Communication Magazine, Accepted (JCR Q1/SCI Q2).
3. **Zhiyuan Qi*(Co-first Author)**, **EMNLP 2026** Anonymous Submission (under review), work done while interning at UIUC, about MLLM Agent.

PATENTS

1. “A Channel Environment Adaptive Joint Optimization Method and System for Sampling-Semantic-Channel Coding”, ZL 2024 1 1509807.8, **Granted**
2. “A Joint Optimization Method, Device, and Medium for Sampling-Semantic in Point Clouds for Holographic Communications”, ZL 2025 1 1141478.0, **Granted**

SELECTED HONORS AND AWARDS

National Scholarship (Awarded to the top0.2% of students nationwide)	2022
Outstanding Undergraduate Thesis Award	2024
Tsinghua University Comprehensive 1st-Class Scholarship	2025
The China Graduate Electronic Design Contest 2nd Prize	2025
MCM (Mathematical Contest In Modeling)/ ICM (Interdisciplinary Contest In Modeling), Finalist Prize (Ranked in the top1% of teams worldwide)	2021

RESEARCH EXPERIENCE

EMNLP 2026 Anonymous Submission (under review), about MLLM Agent. 2026.3 - 2026.5

Research Intern | University of Illinois Urbana-Champaign (UIUC) | Supervisor: [Heng Ji](#) | Champaign, IL in US

A Holographic Classroom Based on Multi-View 3D Reconstruction and Digital Avatar 2025 - present
Student Researcher | Tsinghua University, Department of Electronic Engineering | Beijing, China

- Develop a semantic-driven sampling and compression algorithm for 3D point clouds, which significantly reduces data volume while preserving key information(such as facial features and details), thereby enhancing transmission real-time performance and quality.
- Improve 3D human pose estimation framework and optimize real-time digital human generation through asynchronous pipelining.
- The teacher’s point cloud is captured via 4 Microsoft Azure Kinect DK camera arrays and reconstructed through fusion algorithm. The student’s digital avatar is driven by key facial and limb parameters captured through a monocular RGB camera.

- 10 fps reconstruction frame rate at 200k points/frame scale (200ms latency). Student-side digital human feature stream consumes only 20KB/s bandwidth (100ms latency).

Image Transmission with Joint Sampling and Coding at Adaptive Channel Environments 2024 - 2025
Student Researcher | Tsinghua University, Department of Electronic Engineering | Beijing, China

- Propose a semantic-aware sampling and reconstruction algorithm, dynamically optimize the number of samples in each region of an image. The sampling matrix for each region is individually optimized based on semantic importance, yielding a semantic sampling proportion distribution map shared with the receiver.
- Propose a semantic distribution map-assisted reconstruction algorithm to achieve high fidelity.
- Design an attention-based channel adaptation module to overcome the mismatch between neural network models trained under training channel conditions and those tested under actual channel conditions.
- Achieve reconstruction performance nearly equivalent to that of full data at a 30% sampling rate.
- Noise robustness is enhanced, achieving over 50% sampling rate reduction across diverse channel conditions, outperforming other models across at least a 20dB range of environmental SNR variations.

Subjective Evaluation Perception Dataset for 3D Video Quality 2025 - 2025
Student Researcher | Tsinghua University, Department of Electronic Engineering | Beijing, China

- Based on real-world 3D asset data captured via Microsoft Azure Kinect DK cameras and subjected to 3D reconstruction, incorporating open-resource datasets such as OwlII, 8iVSLF, 8iVFB V2, damage processing including packet loss and noise was applied.
- Participants viewed 3D videos through naked-eye 3D displays and Meta Quest 3 virtual reality headsets, providing perceptual evaluations and ratings across three dimensions: integrity, spatiality and comfort.

AI Psychological Counselor Based on Knowledge Graphs and Emotion Analysis 2022 - 2023
Student Researcher | Beijing JiaoTong University, Department of Electronic Engineering | Beijing, China

- For the professional psychology Q&A module, knowledge graph-driven procedure is achieved through intent recognition through LR+GBDT multi-model fusion and Bert+TextCNN, coupled with entity recognition via BiLSTM-CRF+AC automata. Response quality is enhanced through RL polling strategies.
- For the counselor script generation module, finetune UniLM on targeted datasets across different scenario categories. Based on an enhanced Encoder architecture, it employs a specific self-attention mask mechanism to achieve seq2seq modeling, generating targeted and comforting psychological counseling scripts.

RELATED COURSES

3D Computer Vision (4.0/4.0) | Convex Optimization (4.0/4.0) | Intelligent Wireless Communication (4.0/4.0)
Calculus (4.0/4.0) | Linear Algebra (4.0/4.0) | Probability Theory and Mathematical Statistics (4.0/4.0)
Data Structure (4.0/4.0) | Signal and System (4.0/4.0) | Digital Signal Processing (4.0/4.0)

SKILLS

Python, Pytorch, MATLAB, C/C++, Unity, IsaacLab, Microsoft Azure Kinect DK, Git, Docker, Conda, CMake

SELECTED SERVICES

Reviewer for conferences and journals:

- **2026:** CVPR, IEEE VTC Spring
- **2025:** IEEE ICME, IEEE COMMAG